

Leuschner Large-Area Synoptic Survey (LeuschLASS)

Jun Rui Ting
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We present LeuschLASS, the *Leuschner Large-Area Sky Survey*, a wide-area HI 21 cm survey of the inner Galactic plane carried out with the Leuschner Undergraduate Observatory 4.5 m dish covering $\sim 1.06 \times 10^4 \text{ deg}^2$ of independently-sampled sky at Nyquist sampling for our 3.4° beam, spanning Galactic longitude $\ell \in [-60^\circ, 300^\circ]$ and latitude $|b| \leq 20^\circ$. Each pointing is processed through a seven-stage pipeline: frequency-switching; DC artifact, narrowband RFI and broadband per-dump deviation flagging; LSR frame correction; calibration to $T_{\text{cal}}(t)$ -anchored EBHIS circumpolar reference pointing (diurnal Fourier model $a_0 = 38.4 \text{ K}$, $\pm 13 \text{ K}$ peak-to-peak modulation, 3.35 K RMS). From the calibrated $T_B(\ell, v_{\text{LSR}})$ map and the $|b| \leq 4^\circ$ longitude-velocity strip we recover (i) an inner-Galaxy rotation curve; (ii) an enclosed gravitational mass $M_{\text{grav}}(R_\odot) = (9.6 \pm 0.35) \times 10^{10} M_\odot$ at the IAU anchor; (iii) a strip-confined HI gas mass of $3.2 \times 10^9 M_\odot$, about 3 per cent of the gravitational total; and (iv) a four-arm log-spiral decomposition in $(\ln R, \phi)$ with pitches spanning $i \simeq 5^\circ\text{--}15^\circ$ across clusters B–D, straddling the Sa/Sb morphological boundary. The well-known anomalous-velocity HI features near the Galactic centre (Central Molecular Zone, $\pm 135 \text{ km/s arcs}$, bar streaming) are not cleanly detected.

1. Introduction

The 21 cm hyperfine line of neutral atomic hydrogen is a natural tracer of the underlying structures of the universe. Since HI is optically thin over the bulk of the disk and present at all Galactic latitudes, the line directly traces out where gas sits in (ℓ, b) and how fast it moves along the line of sight.

Crucially, it is also vastly superior to optical tracers for mapping Galactic structure. Interstellar dust extinguishes visible starlight by tens of magnitudes through the disk and renders the inner Galaxy and the far side of the disk essentially opaque at optical wavelengths; at 21 cm the dust optical depth is negligible ($\tau_{\text{dust}} \sim 10^{-7}$) and every sightline through the disk is transparent. As such, the same emission that is blocked from optical view is recovered cleanly in HI.

This same physical signature is also the crux of redshifted 21 cm cosmology: experiments such as PAPER and HERA target the Cosmic-Dawn and Epoch-of-Reionization power spectrum of HI at $z \sim 6\text{--}30$ to probe the first luminous sources and the thermal history of the intergalactic medium (A. R. Parsons et al. 2012). Our focus in this work is the opposite limit to instead characterize $z = 0$ Milky Way Galactic HI emission.

In this report we present **LeuschLASS** (the *Leuschner Large-Area Sky Survey*), a wide-area HI 21 cm survey of the inner Galactic plane carried out with the Leuschner 4.5 m dish. LeuschLASS spans $\sim 1.06 \times 10^4 \text{ deg}^2$ at Nyquist sampling for our 3.4° beam, and we use the calibrated data products to: **(1)** measure the inner-Galaxy rotation curve via the Burton 25-per-cent-of-peak terminal-velocity envelope, and from it derive $M_{\text{grav}}(R)$; **(2)** integrate the column

density across the $b = 0$ strip to obtain an enclosed HI gas mass and compare against the gravitational total; and **(3)** fit a log-spiral model to the locus of bright features in the (ℓ, v_{LSR}) plane and identify any anomalous high-velocity gas near the Galactic centre.

2. Theoretical Background

2.1. HI Brightness Temperatures

The radiative-transfer solution for an isothermal HI cloud of spin temperature T_{spin} and per-frequency optical depth τ_ν is

$$T_B(\nu) = T_{\text{spin}} (1 - e^{-\tau_\nu}). \quad (1)$$

Inverting Eq. (1) for τ_ν and integrating over the line profile, using the HI hyperfine Einstein coefficient to convert $\int \tau_\nu dv$ to column density, yields an exact estimate of the column-density

$$N_{\text{HI}} = 1.823 \times 10^{18} \text{ cm}^{-2} \int T_{\text{spin}} \ln \left[\frac{T_{\text{spin}}}{T_{\text{spin}} - T_B(v)} \right] dv. \quad (2)$$

In the optically thin limit $\tau_\nu \ll 1$ the exponential in Eq. (1) linearises to $T_B \simeq T_{\text{spin}} \tau_\nu$, and the expression simplifies as

$$N_{\text{HI}} = 1.823 \times 10^{18} \text{ cm}^{-2} \int T_B(v) dv \quad [\text{K km s}^{-1}]. \quad (3)$$

With an estimator for an optically thin column density, we can convert our integrated map $W = \int T_B dv$ directly into an enclosed gas mass once a beam solid angle and a distance are specified:

$$M_{\text{gas}}(d) = m_{\text{H}} N_{\text{HI}} \Omega_b d^2, \quad (4)$$

which for the Leuschner 3.4° HPBW Gaussian beam takes

$$\Omega_b = \frac{\pi \theta_b^2}{4 \ln 2} = 4.0 \times 10^{-3} \text{ sr.} \quad (5)$$

For a source larger than the beam, antenna temperature equals brightness temperature, $T_A = T_B$. For an unresolved source $T_A = T_B(\Omega_s/\Omega_b)$. In the case of the inner-Galaxy plane, it substantially overfills the beam at every longitude where we measure the rotation curve. Hence beam dilution is not a major concern in producing the HI maps.

2.2. Frequency Switching

To suppress the channel-dependent receiver gain artifact from using an RTL-SDR (RTL2832U chipset) and isolate the HI emission, we observe by frequency-switching two local-oscillator settings $\text{LO1} = 1419.86 \text{ MHz}$ and $\text{LO2} = 1421.14 \text{ MHz}$ that bracket the rest line $\nu_{\text{HI}} = 1420.4058 \text{ MHz}$. Here, the dimensionless ratio

$$R(\nu) = \frac{I_{\text{LO1}}(\nu) - I_{\text{LO2}}(\nu)}{I_{\text{LO2}}(\nu)} \quad (6)$$

is independent of the receiver gain because gain cancels in numerator and denominator. The system temperature in each LO is then recovered from,

$$T_{\text{sys}}(\nu) = T_{\text{cal}}(t) \frac{P_{\text{off}}(\nu)}{P_{\text{on}}(\nu) - P_{\text{off}}(\nu)}, \quad (7)$$

where P_{on} and P_{off} are the per-channel powers with the front-end noise diode toggled into and out of the signal path.

Rather than adopting the nominal calibrated values for T_{cal} , we determine $T_{\text{cal}}(t)$ empirically by calibrating against a circumpolar reference source revisited throughout observation runs (Section 4.2). With T_{sys} in hand, the absolute brightness scale is recovered as $T_B = R \cdot T_{\text{sys}}$.

2.3. Galactic Kinematics and the Tangent Point

All velocities reported here are Doppler-shifted into the kinematic Local Standard of Rest (LSR), which is an inertial frame independent of the bulk circular motion of nearby disk stars. The conversion from the topocentric frame at the dish to the LSR subtracts the projection of the IAU-standard solar peculiar velocity (apex at $\ell = 56^\circ$, $b = 23^\circ$, $|v| = 20 \text{ km/s}$) onto each pointing,

$$v_{\text{LSR}} = v_{\text{topo}} + \Delta v_{\text{LSR}}(\ell, b, t). \quad (8)$$

For a thin disk in pure circular rotation, the line-of-sight velocity of a voxel at Galactocentric radius R along Galactic longitude ℓ is

$$v_{\text{LSR}}(R, \ell) = \left[\frac{V(R)}{R} - \frac{V_\odot}{R_\odot} \right] R_\odot \sin \ell. \quad (9)$$

Maximising v_{LSR} over the sightline at fixed ℓ gives the terminal velocity v_t , which occurs at the tangent

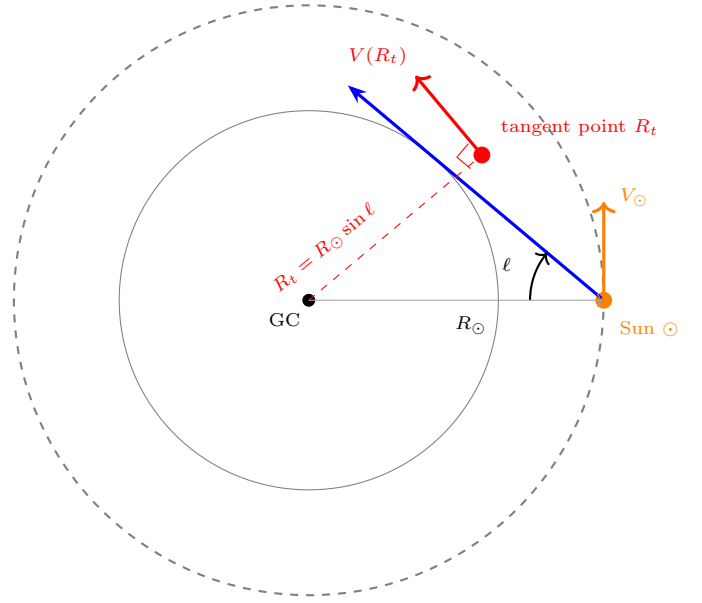


Fig. 1.— Top-down schematic of the inner Galaxy illustrating the tangent-point method. The Sun (orange) orbits at $R_\odot$ with circular speed $V_\odot$; the sightline at Galactic longitude ℓ (blue) reaches its closest approach to the Galactic centre (GC) at the tangent point $R_t = R_\odot \sin \ell$ (red), where the orbital velocity vector $V(R_t)$ is purely radial in the observer's frame. The Doppler velocity at that point is the maximum (Q1) or minimum (Q4) over the entire sightline, $v_t = V(R_t) - V_\odot |\sin \ell|$, so each Q1/Q4 longitude contributes one independent (R_t, V_t) sample of the rotation curve.

point $R_t = R_\odot \sin \ell$ where the sightline is tangent to a circular orbit and the orbital velocity vector is purely radial in the observer's frame. At that point, the line-of-sight velocity is

$$v_t \equiv v_{\text{LSR}}^{\text{max}}(\ell) = V(R_t) - V_\odot |\sin \ell|, \quad (10)$$

and thus the rotation curve at the tangent point is given by,

$$V(R_t) = v_t + V_\odot |\sin \ell|. \quad (11)$$

We adopt as priors the nominal values $R_\odot = 8.5 \text{ kpc}$ and $V_\odot = 220 \text{ km/s}$ (F. J. Kerr & D. Lynden-Bell 1986), and identify v_t with the 25-per-cent-of-peak envelope position of the spectrum at each ℓ following W. B. Burton (1988). With $V(R)$, the enclosed gravitational mass follows by a spherical-shell system,

$$M_{\text{grav}}(R) = \frac{V^2(R) R}{G}. \quad (12)$$

For a sightline that is not at the tangent point, the inversion from v_{LSR} to heliocentric distance is degenerate in that the sightline crosses the heliocentric distance twice with solutions parameterized as

$$R^2 = R_\odot^2 + d^2 - 2R_\odot d \cos \ell, \quad (13)$$

$$d_\pm(R, \ell) = R_\odot \cos \ell \pm \sqrt{R^2 - R_\odot^2 \sin^2 \ell}. \quad (14)$$

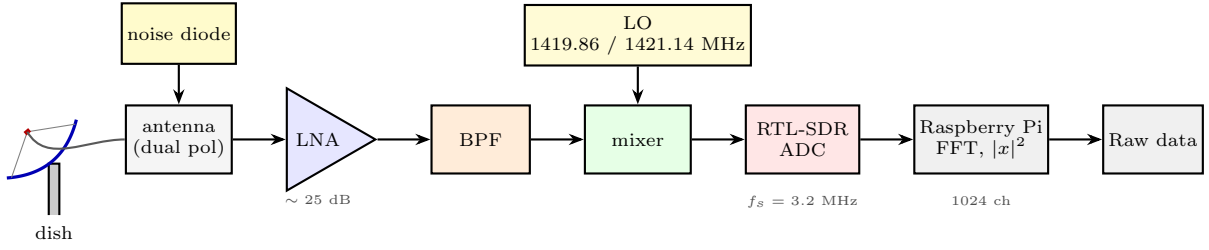


Fig. 2.— Physical RF chain from feed to disk. The noise diode is switched on for the “cal” subset of dumps; the LO is switched between the two values 1419.86 and 1421.14 MHz on the ABBA cadence to bracket the rest line for frequency-switching subtraction. One identical chain runs per polarisation; the autocorrelated powers `corr00` and `corr11` are summed at the spectral-ratio level to recover Stokes I .

For $R_t < R < R_\odot$ both roots are real and positive, so each (ℓ, v_{LSR}) pixel maps to two physical heliocentric distances: the near branch d_- (in front of the tangent point); and the far branch d_+ (after it). For $R > R_\odot$ only d_+ is positive and the ambiguity disappears. Breaking the degeneracy at a generic pixel requires external information (e.g., HI self-absorption silhouettes, parallax maser distances, or extinction-mapped dust associations) that is unavailable in our single-dish data, so we report both branches separately for the gas-mass integral in Section 5.

3. Instrument and Observations

3.1. Equipment

The data presented here were taken with the Leuschner Undergraduate Observatory 4.5 m parabolic dish, located at $\phi_{\text{obs}} = 37.9183^\circ \text{ N}$, $\lambda_{\text{obs}} = -122.1067^\circ \text{ E}$, elevation 304 m above sea level in Berkeley, California. The dish’s half-power beamwidth at 1.42 GHz is $\theta_b \simeq 3.4^\circ$, equivalent to a beam solid angle $\Omega_b = 4.0 \times 10^{-3} \text{ sr}$ under a Gaussian-beam approxi-

mation. A two-polarisation antenna captures the sky power, fed into bandpass filters and low-noise amplifiers, after which each polarisation chain is mixed with a local oscillator and digitised at 3.2 MHz by an RTL-SDR. The digitiser hands a 2^{14} -sample block to a Raspberry Pi, which performs a 1024-channel FFT of the autocorrelation power; 1025 such blocks per dump are averaged to yield a 5.25 s integration. The full RF chain is shown in Figure 2.

3.2. Observation Plan

The instrument is driven by a forward-simulating scheduler that tessellates the accessible sky into a brick-pattern grid of $2^\circ \times 2^\circ$ cells, with longitudinal pitch corrected for foreshortening via $\Delta\ell = 2^\circ / \cos b$. The horizontal pattern is split into two latitude phases, $b \in \{0, \pm 2, \pm 4, \pm 6\}^\circ$ (even phase) and $b \in \{\pm 1, \pm 3, \pm 5\}^\circ$ (odd phase, displaced by half a step in ℓ), so that the resulting nearest-neighbour separation is $\sqrt{2}^\circ$, Nyquist-sampled for the 3.4° beam. Cells are scanned along longitudes with zig-zag in b , minimising slew distance and north-polar azimuth-wrap

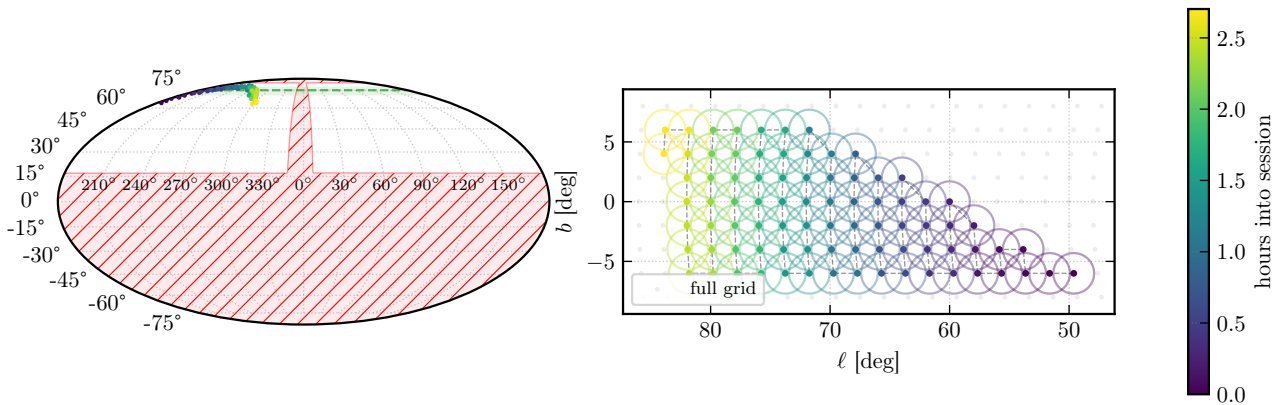


Fig. 3.— Per-session coverage of a survey session, viewed in both the topocentric mount frame (left, Mollweide of (az, alt) with the never-observable wedge hatched) and Galactic flat-sky (right, (ℓ, b) with the full Nyquist grid as grey background and beam ellipses $\cos b$ -stretched in longitude). Scatter is coloured by hours into the session; the dashed line traces the ascending longitude zig-zag constant latitude scan order. The green band and dashed line on the topocentric panel mark the interquartile range and median of the science-pointing altitudes; the planner deliberately concentrates pointings near the upper accessibility limit to suppress ground pickup and air-mass excess, with 100 per cent of pointings in this session falling at $\text{alt} \geq 60^\circ$.

excursions. The Leuschner mount has hard limits $17^\circ \leq \text{alt} \leq 83^\circ$ and $7^\circ \leq \text{az} \leq 348^\circ$, and the resulting accessibility wedge in (ℓ, b) is precomputed at each session start; forecasted cells outside the wedge are skipped.

Each cell is observed with an ABBA-interleaved (A=LO1, B=LO2) schedule of 4 cal dumps and 8 obs dumps spanning both LOs, giving four surviving LO1/LO2 obs pairs per cell. A periodic recalibration pointing is injected every 10 science cells at a fixed position, $(\text{RA}, \text{Dec}) = (90^\circ, +72^\circ) \rightarrow (\ell, b) = (141.9^\circ, +21.9^\circ)$, anchored to the Effelsberg-Bonn HI Survey (EBHIS) reference brightness (B. Winkel et al. 2016); the recal injection costs ~ 24 per cent of telescope time but pins the noise-diode amplitude against thermal drift. Across 64 sessions we accumulated 2709 (session, cell) calibration scalars covering $\sim 1.06 \times 10^4 \text{ deg}^2$ of unique science sky (foreshortening-corrected) in $\ell \in [-60^\circ, 300^\circ]$ and $|b| \leq 20^\circ$. Figure 3 depicts as an example the longest contiguous archived session in both the topocentric and Galactic frames. The time-coloured trace illustrates the zig-zag scan order, the $\cos b$ longitude foreshortening of the brick pattern, and the recalibration excursions to the EBHIS anchor.

4. Data Reduction and Calibration

The reduction pipeline takes raw per-dump `.npz` autocorrelations and produces a calibrated brightness-temperature spectrum $T_B(\nu)$ per cell in seven ordered stages:

1. **Data ingest.** Parse per-session `.npz` dumps and apply Sun/Moon exclusion ($> 30^\circ$ from the Sun, $> 10^\circ$ from the Moon).
2. **Per-dump flagging.** FFT-shift each 1024-channel autocorrelation and mask the DC bin (channel 512); run a sliding-window Chebyshev pseudo-continuum fit (degree 3, kernel 15) and reject channels deviating by more than 10σ of the per-spectrum MAD; then reject whole dumps whose channel-wise residuals exceed 5σ on > 5 per cent of channels.
3. **Frequency-switched ratio.** Form $R(\nu)$ of Eq. (6) dump-by-dump and average surviving LO1/LO2 pairs within each cell.
4. **LSR shift.** Shift the averaged spectrum to the LSR frame using Eq. (8).
5. **Cross-session QA.** Apply a 5σ MAD filter to cross-session pairs (reject pairs with > 15 per cent deviant channels) and flag cells with < 3 surviving pairs into a reobserve list.
6. **$T_{\text{cal}}(t)$ drift solve.** At every recal-pointing visit, pair the Leuschner ratio $R_{\text{pol}1}$ with the EBHIS reference brightness via a two-peak differential to back out a per-visit $T_{\text{cal}, \text{pol}1}$ from Eq. (7); fit the 192 anchored values with a $K = 1$ diurnal Fourier model $T_{\text{cal}}(h) = a_0 +$

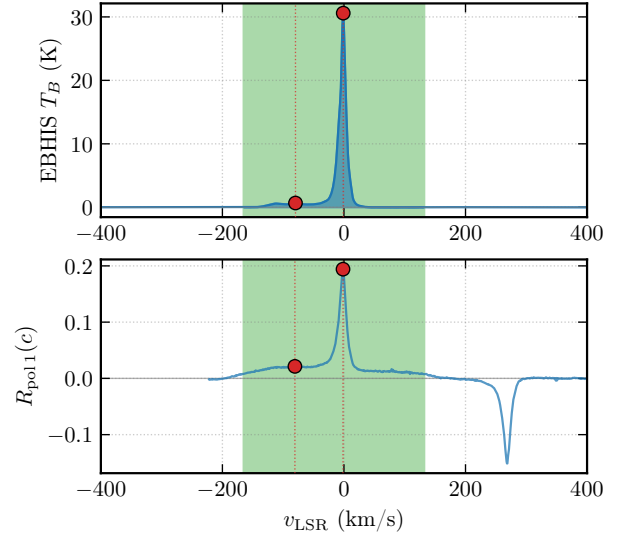


Fig. 4.— Two-peak differential-peak anchor at the recal pointing $(\ell, b) = (141.9^\circ, +21.9^\circ)$. *Top*: EBHIS reference $T_B(v_{\text{LSR}})$ with the Leuschner v_{LSR} window shaded green; the two anchored peaks are marked in red. *Bottom*: Leuschner frequency-switched ratio $R_{\text{pol}1}(\nu)$ averaged across all visits to the same pointing, with the matching peaks marked. Pairing T_B^{EBHIS} with $R_{\text{pol}1}$ at each peak gives one independent T_{cal} measurement per visit.

$a_1 \cos(2\pi h/24) + b_1 \sin(2\pi h/24)$, and evaluate at each science cell's observation time.

7. **Brightness recovery.** Construct $G(\nu) = (P_{\text{on}}(\nu) - P_{\text{off}}(\nu))/T_{\text{cal}}(t)$, smooth by a Chebyshev fit (degree 2, one-sided 2σ clip to preserve HI emission), and recover $T_{\text{sys}, \text{pol}1}(\nu)$ per cell via Eq. (7). Combine with the dual-polarisation ratio to give the Stokes- I brightness $T_B(\nu) = 2 R_{\text{pol}1}(\nu) T_{\text{sys}, \text{pol}1}(\nu)$.

4.1. RFI Flagging and Outlier Rejection

Stage 2 of the pipeline screens each raw 1024-channel autocorrelation for instrumental and environmental contaminants before the dump enters the frequency-switched ratio. Next, we perform a 2-step outlier rejection, stage 2 which checks for per-channel RFI spikes, and stage 5 which checks for broadband deviations.

First, a sliding-window Chebyshev pseudo-continuum is fit to each spectrum (polynomial degree 3, kernel width 15 channels), local extrema are excluded from the fit so that real emission spikes do not bias the baseline, and the per-spectrum residual MAD is computed. Channels deviating by more than 10σ of the MAD from the pseudo-continuum are flagged out. The Chebyshev choice over a single-degree polynomial captures the gentle bandpass curvature without overfitting, and the 10σ threshold is conservative enough that the strongest HI emission ($T_B \sim 100$ K, well

above any plausible RFI in the protected 1420 MHz band excluding LO leakage) is never rejected; in practice fewer than 1 per cent of channels per spectrum are flagged.

A final pass catches whole-dump glitches that escape the per-channel cut (e.g., thermal excursions) that distort the entire bandpass rather than a single channel. For each (session, cell, LO) group we fit a per-dump shape template, compute channel-wise residuals against it, and reject any dump for which > 5 per cent of channels deviate by more than 5σ of the per-dump MAD. This integrates over the spectrum rather than thresholding it pixel-by-pixel, so it is sensitive to broadband shape distortions that a per-channel filter would miss. The surviving dumps are passed to stage 3 for ratio formation.

4.2. Noise-Diode Drift Calibration

The absolute brightness scale is set by the noise-diode amplitude T_{cal} , with nominal calibrated values (79 K for polarisation 1, 58 K for polarisation 0). However, due to possible equipment fault, and more pertinently ambient temperature and front-end loading vary between morning and afternoon sessions, T_{cal} drifts at the ± 15 per cent level over a 24 h cycle. As such, it necessitates additional calibration steps. Stages 6–7 of the reduction pipeline solve for this drift empirically by anchoring to EBHIS, and propagate it into the final brightness map.

The anchor uses a two-peak differential-peak method. The EBHIS reference spectrum at the recal pointing has one well-isolated emission peak at $v_{\text{LSR}} \sim 0$ and a fairly flat emission plateau at $v_{\text{LSR}} < 0$, separated by > 50 km/s, and the Leuschner ratio spectrum $R_{\text{pol}1}(v)$ contains the same two features (Figure 4). Pairing the EBHIS brightness $T_{B,i}^{\text{EBHIS}}$ at each peak with the corresponding Leuschner ratio R_i gives two simultaneous equations $T_{B,i}^{\text{EBHIS}} = (T_{\text{sys}} + T_{\text{cal}}) R_i$ for $i \in \{1, 2\}$ that solve for $T_{\text{cal}, \text{pol}1}$ at that visit via Eq. (7). Only pol-1 is used to set the absolute brightness temperature scale. The pol-0 noise-diode cou-

pling at the 3.2 MHz sample rate is nonlinear and the implied $T_{\text{cal}, \text{pol}0}$ drifts to unphysical values, so we exclude it from the anchor.

The 192 anchored pol-1 visits across 64 sessions are then fitted with a 24 h PDT Fourier model, $T_{\text{cal}}(h) = a_0 + a_1 \cos(2\pi h/24) + b_1 \sin(2\pi h/24)$, with a single diurnal harmonic ($K = 1$). Figure 5 shows the raw per-visit $T_{\text{cal}, \text{pol}1}(t)$ scatter coloured by altitude; the diurnal envelope is clearly visible, and the median across all visits is 41.7 K. Figure 6 folds the same data into a 24 h phase with the $K = 1$ Fourier fit overlaid, yielding $a_0 = 38.4$ K with a ± 13 K peak-to-peak modulation and a fit RMS of 3.35 K. The constant term is roughly half the nominal 79 K plate value.

With $T_{\text{cal}}(t)$ now defined at every observation time, the per-cell gain spectrum is constructed as $G(\nu) = (P_{\text{on}}(\nu) - P_{\text{off}}(\nu))/T_{\text{cal}}(t)$, smoothed by a Chebyshev fit (degree 2, one-sided 2σ clip to preserve the HI emission), and the resulting per-cell $T_{\text{sys}, \text{pol}1}(\nu)$ is combined with the dual-polarisation ratio $R_{\text{pol}1}$,

$$T_B(\nu) = 2 R_{\text{pol}1}(\nu) T_{\text{sys}, \text{pol}1}(\nu),$$

to give the calibrated brightness scale per cell. The factor of two preserves the unpolarised-source identity $I = 2 T_{B, \text{pol}1}$; $R_{\text{pol}1}$ retains both polarisation chains at the ratio level (Stokes $I = \text{corr00} + \text{corr11}$) because they are independent radiometric realisations of the same sky signal. Here, we assume that the sources are unpolarised as thus a calibrated polarisation 0 and 1 are identical.

4.3. Calibrated Maps

The first science-grade data product is the gridded all-sky map of integrated brightness $W(\ell, b)$ shown in Figure 7. Cell samples are first deposited onto the sinusoidal projection

$$u = (\ell - \ell_0) \cos b, \quad v = b, \quad (15)$$

which is equal-area and absorbs the $\cos b$ foreshortening, so each cell carries its true solid-angle weight

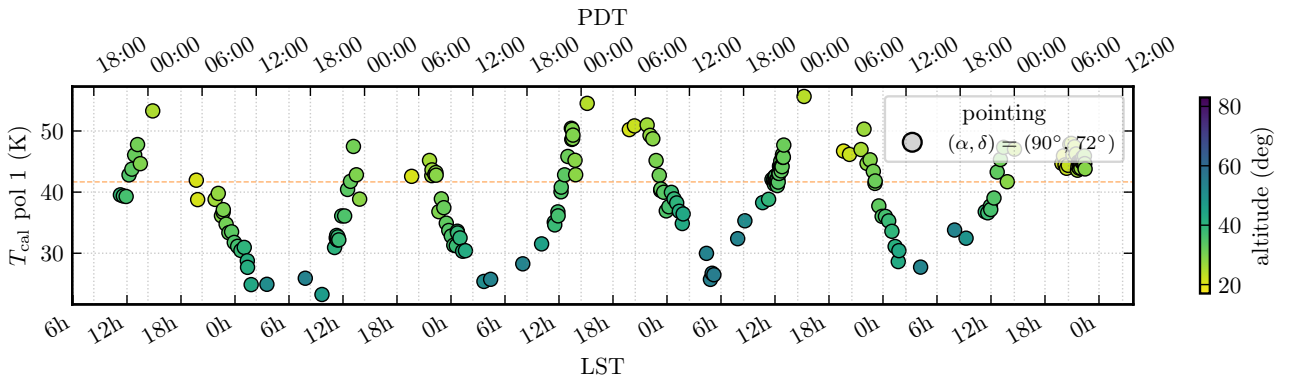


Fig. 5.— Per-visit $T_{\text{cal}, \text{pol}1}(t)$ across the 4.5 day calibration window, scatter coloured by pointing altitude. The diurnal envelope is visible by eye; the dashed line is the across-visits median. Sessions are shaded at the top axis with LST and PDT clock dials.

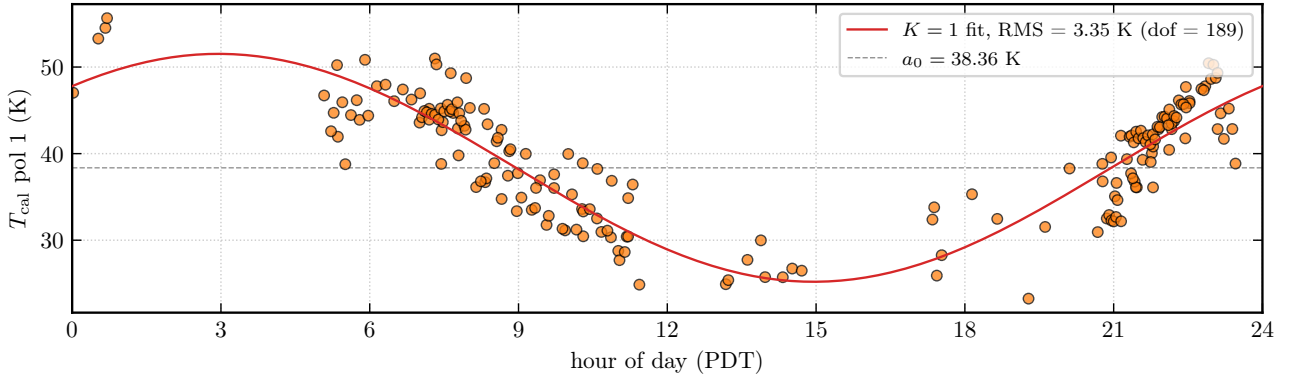


Fig. 6.— 24 h PDT fold of $T_{\text{cal, pol 1}}(t)$ with the $K = 1$ Fourier fit (red); the constant term $a_0 = 38.4$ K is shown dashed.

without an explicit Jacobian. The Leuschner beam is well approximated by a stationary 2-D Gaussian in the (u, v) frame,

$$K(\Delta u, \Delta v) = \frac{1}{2\pi\sigma^2} \exp\left[-\frac{\Delta u^2 + \Delta v^2}{2\sigma^2}\right], \quad (16)$$

$$\sigma = \frac{\theta_b}{2\sqrt{2\ln 2}} \simeq 1.44^\circ, \quad (17)$$

truncated at 2σ for numerical efficiency. The map is then formed by the convolution

$$\begin{aligned} \widetilde{W}(u, v) &= (W \star K)(u, v) \\ &= \iint W(u', v') K(u - u', v - v') du' dv', \end{aligned} \quad (18)$$

evaluated in Fourier space as $\widehat{\widetilde{W}} = \widehat{W} \cdot \widehat{K}$ via FFT, and resampled to 0.5° pixels in (ℓ, b) . The planar convolution (18) is functionally identical to a great-circle Gaussian smoothing on the sphere.

The bright equatorial ridge at $|b| < 5^\circ$ traces the cold neutral disk; off-plane gas is detected at $W \sim 200\text{--}500$ K km/s down to $|b| \sim 15^\circ$, demonstrating that the EBHIS-anchored calibration recovers diffuse high-latitude HI at the same flux scale as the LAB and HI4PI surveys.

The second science product is the calibrated longitude–velocity strip at $b \sim 0$, shown later in Figure 11 together with the spiral-arm overlay, beam-weighted into 0.5° longitude pixels across cells with $|b| \leq 4^\circ$.

5. Galactic Dynamics from the Tangent Point

5.1. Rotation Curve

We extract the terminal-velocity envelope at each longitude in $\ell \in [20^\circ, 65^\circ]$ following the W. B. Burton (1988) 25-per-cent-of-peak rule where the per-spectrum peak $T_{B, \text{max}}(\ell)$ is identified within the edge-safe velocity band $|v_{\text{LSR}}| < 130$ km/s, and the terminal velocity $v_t(\ell)$ is the largest velocity at which

$T_B > 0.25 T_{B, \text{max}}$. The 25-per-cent threshold sits well above the off-line column noise ($\sigma_{T_B} \simeq 0.3$ K, median over ℓ) but well below the bright knee of the spectrum, where shot-noise scatter on a sharp envelope edge would dominate. We restrict to $\ell \geq 20^\circ$ to avoid the inner-degree region where circular rotation breaks down (bar-driven streaming, RFI contamination) and to $\ell \leq 65^\circ$ where the tangent point is still well-defined at our beam.

Inverting Eq. (11) on the 46 surviving longitudes yields the rotation curve plotted in Figure 8 alongside the empirical J. Brand & L. Blitz (1993) parameterisation. Inside $R \sim 4$ kpc the measured curve falls roughly 20 km/s below BB93 where the fractional-peak treatment biases low when the line profile is heavily overlapped along the sightline; in $R = 4\text{--}6$ kpc the agreement is at the 5-per-cent level. Between $R \simeq 6\text{--}7$ kpc the measured curve sits $\sim 10\text{--}15$ km/s above BB93, consistent with the systematic outward streaming residual of 5–10 km/s from the Sagittarius–Carina arm tangents at $\ell \simeq 45^\circ\text{--}55^\circ$ (M. J. Reid et al. 2014).

Three independent sources of noise enter the recovered $V(R_t)$: channel resolution $\delta v_{\text{ch}} = c \Delta f_{\text{ch}} / \nu_{\text{HI}} = 0.66$ km/s; photon noise $\sigma_{v_t}^{\text{rad}} \simeq 0.06$ km/s; and calibration drift $\sigma_{v_t}^{\text{cal}} \simeq 1.8$ km/s, dominated by the 3.35 K $K = 1$ Fourier-fit residual on $T_{\text{cal, pol 1}}$ (Section 4.2).

Quadrature-summing gives $\sigma_{v_t}^{\text{stat}} \simeq 1.8$ km/s, but the across- ℓ scatter of $V(R_t)$ about a smoothed monotone fit is larger ($\simeq 4$ km/s), implying a residual systematic floor of $\simeq 3.6$ km/s from streaming motions and beam-overlapping. We adopt the empirical $1\text{-}\sigma$ value $\sigma_{v_t} = 4$ km/s per longitude, propagated through Eq. (10) ($|dV/dv_t| = 1$) to produce the shaded band in Figure 8.

5.2. Enclosed Gravitational and Gaseous Mass

Substituting $V(R)$ into Eq. (12) yields the enclosed gravitational mass profile plotted in Figure 9. The IAU anchor at $R_\odot$ corresponds to $M_{\text{grav}}(R_\odot) = V_\odot^2 R_\odot / G = 9.6 \times 10^{10} M_\odot$, and the measured curve reaches this value smoothly. Propagating $\sigma_{v_t} = 4$ km/s through Eq. (12) gives $\sigma_{M_{\text{grav}}}(R_\odot) \simeq 3.5 \times$

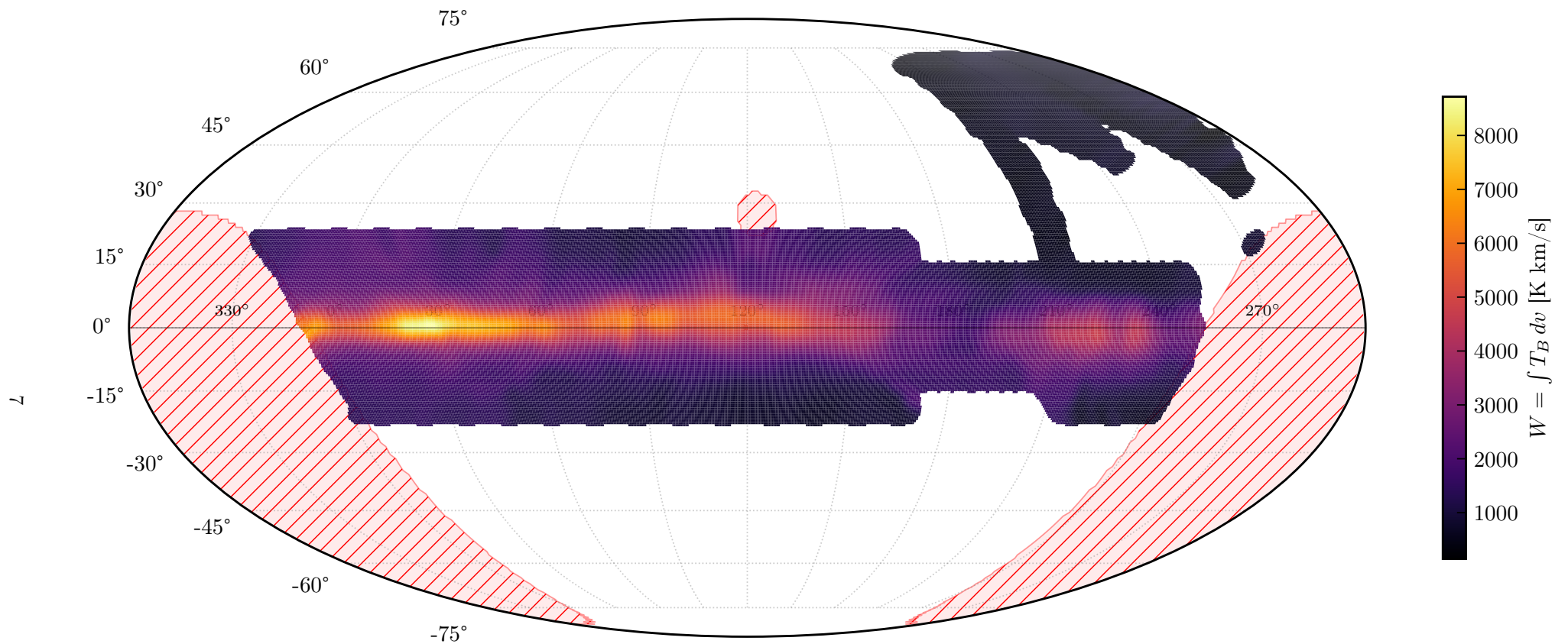


Fig. 7.— Beam-weighted gridded map of integrated HI brightness $W = \int T_B dv$ in K km/s, centred on $\ell = 120^\circ$ in Mollweide projection. Samples are deposited onto a sinusoidal (u, v) grid via $u = (\ell - \ell_0) \cos b$, $v = b$, convolved with a Gaussian kernel of FWHM 3.4° (HPBW) truncated at 2σ , and resampled to 0.5° pixels. The red-hatched wedge is the patch of sky outside the Leuschner accessibility envelope $17^\circ \leq \text{alt} \leq 83^\circ$ and is never observed. The bright equatorial ridge at $|b| < 5^\circ$ traces the cold neutral disk, with the Cygnus–Outer arm complex visible near $\ell \sim 80^\circ$ and the inner-disk peak $W \sim 2.5 \times 10^3$ K km/s saturating near $\ell \sim 30^\circ$.

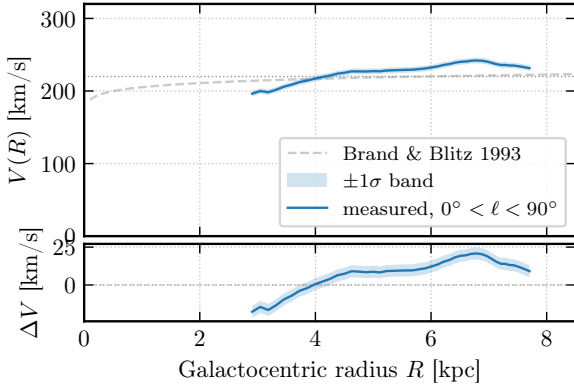


Fig. 8.— Inner-Galaxy rotation curve $V(R_t)$ from the Burton 25-per-cent-of-peak tangent-velocity rule on the $|b| \leq 4^\circ$ strip (blue solid). The shaded blue band is the empirical $\pm 1\sigma$ uncertainty propagated from $\sigma_{v_t} = 4$ km/s per longitude. The dashed grey curve is the J. Brand & L. Blitz (1993) empirical parameterisation $V(R) = V_\odot [a_1(R/R_\odot)^{a_2} + a_3]$ for comparison; the dotted grey horizontal line marks the IAU anchor $V_\odot = 220$ km/s, and the dotted grey vertical line marks $R_\odot = 8.5$ kpc. The bottom panel shows the residual $\Delta V = V_{\text{meas}} - V_{\text{BB93}}$ with the same uncertainty band.

$10^9 M_\odot$ ($\simeq 3.6$ per cent), shown as the shaded band in Figure 9.

Figure 10 plots the cumulative HI gas mass derived from per-pixel summation of Eqs. (3)–(4) along the $|b| \leq 4^\circ$ strip on a 2-degree longitude grid, separately for the near and far branches of the kinematic distance ambiguity.

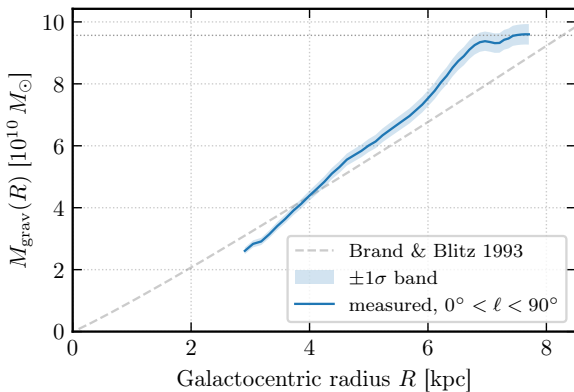


Fig. 9.— Enclosed gravitational mass profile $M_{\text{grav}}(R) = V^2 R / G$ on the measured rotation curve (blue solid). The shaded blue band is the propagated $\pm 1\sigma$ uncertainty $\sigma_M/M = 2\sigma_V/V$ from $\sigma_{v_t} = 4$ km/s. The dashed grey curve is the same inversion applied to the Brand & Blitz 1993 rotation curve, and the dotted grey horizontal line marks the IAU anchor $M_{\text{grav}}(R_\odot) = V_\odot^2 R_\odot / G \simeq 9.6 \times 10^{10} M_\odot$ (vertical dotted line at $R_\odot = 8.5$ kpc).

matic distance. The strip-confined gas mass totals $3.1 \times 10^8 M_\odot$ in the near branch and $3.2 \times 10^9 M_\odot$ in the far branch, i.e. between 0.3 per cent and 3.4 per cent of the gravitational total. The near/far ratio approximately matches the geometric factor d^2 , indicating that the bulk of the HI emissivity sits at larger heliocentric distance. The total HI mass of the Galaxy is closer to $8 \times 10^9 M_\odot$ (P. M. W. Kalberla & J. Kerp 2009) which suggests that our underestimate is the expected consequence of restricting the integration to the $b = 0$ strip (the warp and disk thickening above $|b| > 4^\circ$ are not captured) and of assuming optical thinness in the bright inner plane where self-absorption is known to remove ~ 20 per cent of the column density (J. M. Dickey & F. J. Lockman 1990).

6. Spiral Structure

The $T_B(\ell, v)$ strip is rich with discrete bright tracks that connect smoothly across many tens of degrees of longitude which suggests the presence of longitudinally-extended spiral arms. We isolate the spine of every arm by running a 1-D peak finder on each longitude column of the strip, and invert each (ℓ, v) peak to a Galactocentric (R, ϕ) pair using the full-disk extension of the measured rotation curve (flat outside $R_\odot$). We then run a 4-means clustering in standardised $(\ln R, \phi)$ space, label the clusters A–D in order of increasing mean R , and fit a log-spiral $\ln R = \ln R_0 + \kappa \phi$ to each cluster by linear least squares; the pitch angle $i = \arctan \kappa$ is a characteristic angle between the spiral arm and the local Galactocentric circle at every radius. Cluster A samples the inner-Galaxy ridge contaminated by the inner-degree RFI and therefore yields a steeper, less physically interpretable slope; clusters B–D recover

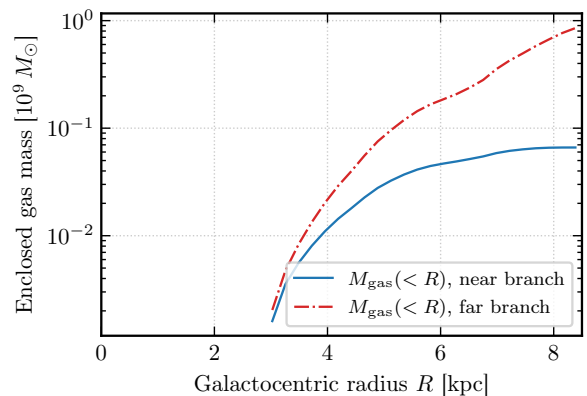


Fig. 10.— Cumulative HI gas mass $M_{\text{gas}}(< R)$ on the $|b| \leq 4^\circ$ strip, near branch (blue) and far branch (red dot-dashed) of the kinematic distance ambiguity. The far branch dominates by an order of magnitude because $M_{\text{gas}} \propto d^2$ at fixed T_B , and the bulk of HI emissivity sits well beyond the Sun on Q1 sightlines.

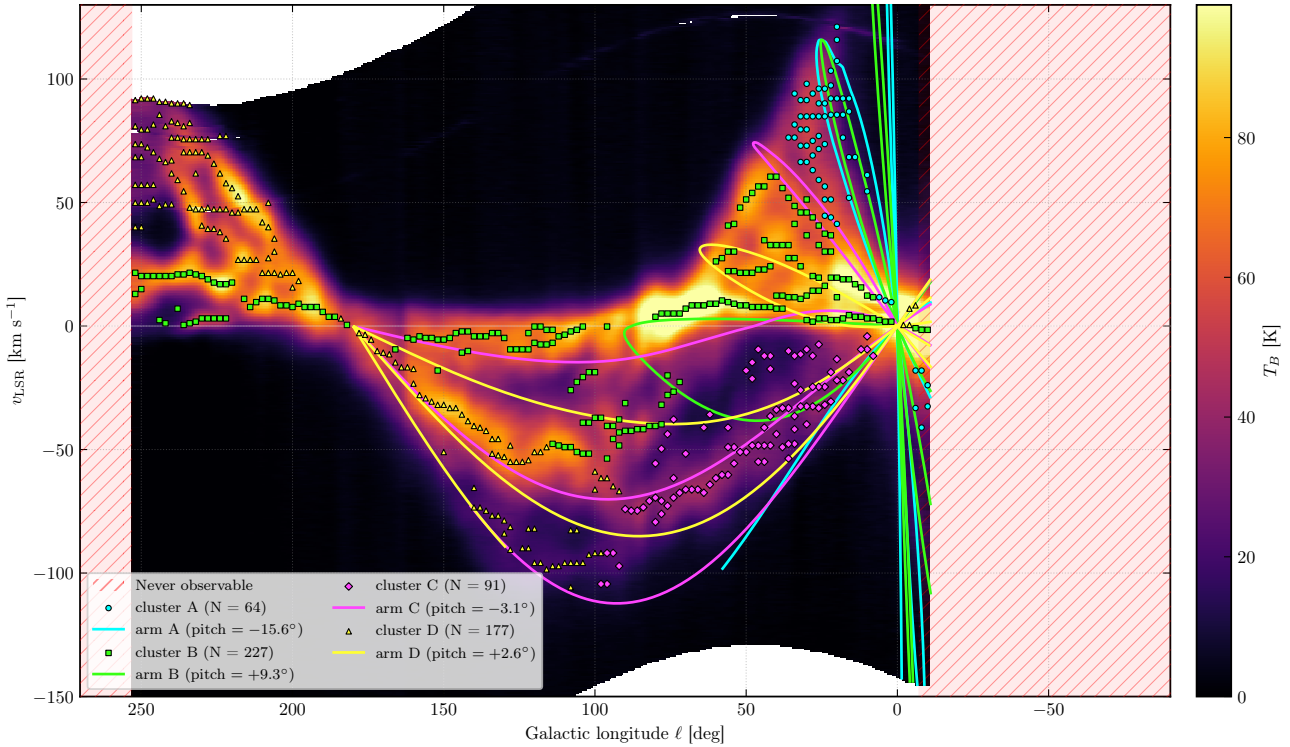


Fig. 11.— 4-arm K-means log-spiral fit overlaid on the $T_B(\ell, v_{\text{LSR}})$ strip at $b \sim 0$. Markers are isolated local maxima of $T_B(v)$ at each surveyed longitude, partitioned by 4-means in standardised $(\ln R, \phi)$ space. Solid curves are the forward-projection of the per-cluster log-spiral fit through the full-disk rotation curve.

pitches in the range expected for a multi-arm Milky Way (M. J. Reid et al. 2014).

Figure 11 shows the four-arm K-means partition and its associated log-spiral fits in the (ℓ, v) plane. The pitch angles printed in the legend span roughly $5\text{--}15^\circ$ across clusters B, C, and D, comfortably within the parallax-anchored M. J. Reid et al. (2014) multi-arm pitch range. However, we cannot tag individual clusters to specific named arms (Sagittarius–Carina, Perseus, Norma, Outer, Scutum–Centaurus) without parallax distances.

Figure 12 re-projects the same data into the Galactocentric (x, y) plane through the kinematic-distance inversion (near branch inside $R_\odot$, far branch outside) and overlays all four fitted spirals in their natural top-down geometry; the spiral structure of the inner-disk HI ridge is plainly visible, and the bulge-dominated central concentration is the bright unresolved region at the origin.

The four-arm decomposition resolves a clear two-tier primary and secondary arm structures. Clusters B and D track densely populated bright ridges we identify as the two primary arms of the partition, while clusters A and C trace sparser, less bright ridges that we identify as secondary arms following the dominant pair at a smaller offset in $(\ln R, \phi)$.

The arms are, however, not cleanly resolved since in both the (ℓ, v) and the top-down $(x_{\text{gc}}, y_{\text{gc}})$ projections (Figures 11 and 12) the secondary arms

overlap the primaries over much of their length. We attribute this to a combination of factors: (i) the limited longitude coverage as the $|\ell| > 120^\circ$ Southern Galaxy is below the Leuschner horizon and leaves several arms tangentially under-sampled, particularly toward the far inner Galaxy where the most kinematically distinct tracks would lie; (ii) the kinematic-distance ambiguity (Eq. (14)) doubles each (ℓ, v) pixel into two physical positions without an independent way to break the degeneracy at the spiral-arm level, blurring the top-down inversion; and (iii) the 3.4° HPBW corresponds to several hundred parsecs along the line of sight in the inner Galaxy, beam-blending parallel arm segments that fall within the same beam. Robust arm identification at the parallax-level cleanliness of M. J. Reid et al. (2014) would require deeper sensitivity at high $|b|$ together with external distance priors (e.g., maser parallaxes or HI self-absorption profiles) to break the aforementioned degeneracy.

7. Discussion

7.1. Galaxy Morphology

The Hubble–de Vaucouleurs morphological sequence binds spiral pitch angle to bulge-to-disk ratio where of note, Sa galaxies are tightly wound ($i \simeq 5^\circ$, large bulges), Sb intermediate ($i \simeq 10\text{--}15^\circ$), and Sc loosely wound ($i \simeq 15\text{--}30^\circ$, small bulges) (J. Binney & M. Merrifield 1998). Our four-arm K-means de-

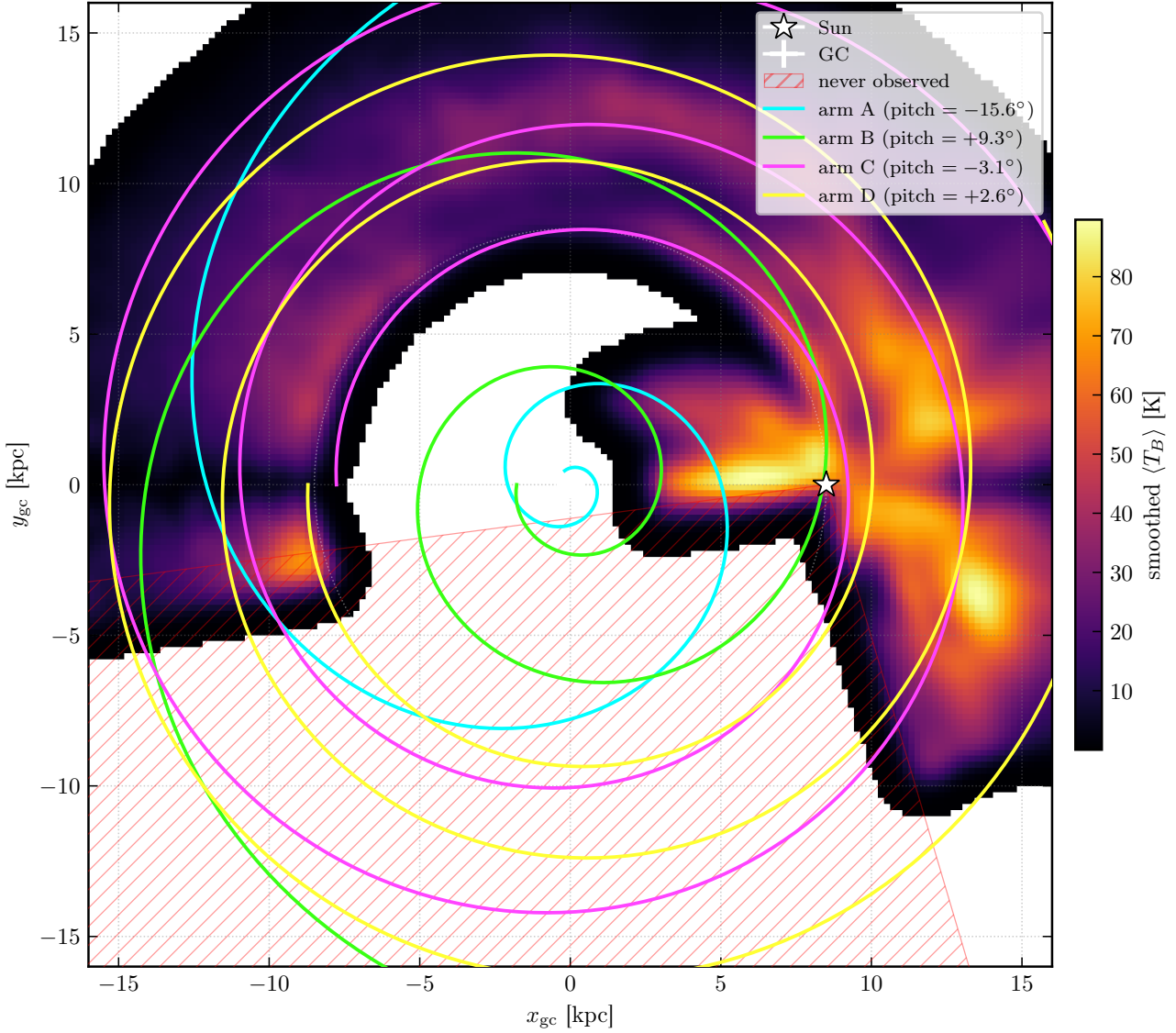


Fig. 12.— Top-down kinematic-distance projection of $T_B(\ell, v_{\text{LSR}})$ into the Galactocentric $(x_{\text{gc}}, y_{\text{gc}})$ plane (near branch inside $R_\odot$, far branch outside). The Sun ($\star$) sits at $(8.5, 0)$ kpc and the Galactic centre ($+$) at the origin; the dotted circle is the Solar circle. The fitted log spirals are overlaid as solid curves traced over multiple wraps.

composition recovers pitches in the range $i \simeq 5^\circ\text{--}15^\circ$ across clusters B–D, straddling the Sa/Sb boundary and consistent with the classification of the Milky Way as an SBb spiral and with the parallax-anchored M. J. Reid et al. (2014) multi-arm pitch range of $9\text{--}13^\circ$.

From previous literature, we expect to observe a well-known anomalous high-velocity CMZ envelope spanning $-200 \lesssim v_{\text{LSR}} \lesssim +200$ km/s (H. S. Liszt 2008), the ± 135 km/s arcs of W. B. Burton (1971), and the bar-driven streaming motions of J. Binney et al. (1991). However, they are not cleanly detected in our $|\ell| < 5^\circ$ columns. The bright narrow feature near $|v_{\text{LSR}}| \simeq 115$ km/s is identified as RFI, and the genuine high-velocity gas at $T_B \sim 1\text{--}5$ K (HI4PI / EBHIS) sits at or below our per-cell off-line noise floor $\sigma_{T_B} \simeq 0.3$ K after beam dilution by the ~ 540 pc Leuschner footprint at the GC dis-

tance (a factor $(540/200)^2 \sim 7$ for the $\lesssim 200$ pc CMZ). As such, an SMBH-mass estimate is therefore not attempted. Notably, even if the CMZ envelope had been recovered, $M = v^2 r / G$ would return the integrated bulge mass within the ~ 540 pc beam ($\sim 10^{10} M_\odot$) rather than Sgr A*’s $4.30 \times 10^6 M_\odot$ (A. M. Ghez et al. 2008), and the underlying gas would likely be on bar-driven streaming trajectories rather than Keplerian orbits around the black hole.

8. Conclusions

LeuschLASS demonstrates that a single-dish, large-area HI 21 cm survey can recover the bulk dynamics, gas content, and spiral morphology of the inner Galaxy when paired with a careful EBHIS-anchored absolute-brightness chain. From $\sim 1.06 \times 10^4 \text{ deg}^2$ of fully Nyquist-sampled sky in

$\ell \in [-60^\circ, 300^\circ]$, $|b| \leq 20^\circ$ we obtain four primary results.

The Burton 25-per-cent-of-peak tangent-velocity rule applied to the $|b| \leq 4^\circ$ strip yields $V(R_t)$ in agreement with J. Brand & L. Blitz (1993) at the 5-per-cent level over $R \simeq 4\text{--}8$ kpc, with the inner-disk low bias and the $R \simeq 6\text{--}7$ kpc high bias understood as the standard fractional-of-peak blending artefact and Sagittarius–Carina arm streaming respectively. Additionally, this gives an enclosed gravitational mass $M_{\text{grav}}(R_\odot) = (9.6 \pm 0.35) \times 10^{10} M_\odot$ at the IAU anchor. Optically-thin per-pixel summation over the $b = 0$ strip yields a far-branch HI gas mass of $3.2 \times 10^9 M_\odot$ (or $3.1 \times 10^8 M_\odot$ on the near branch) (i.e. 0.3–3.4 per cent of the gravitational total) where the residual to the literature value of $\sim 8 \times 10^9 M_\odot$ (P. M. W. Kalberla & J. Kerp 2009) is attributable to the $|b| > 4^\circ$ exclusion and to the ~ 20 per cent optically-thin underestimate in the bright inner plane. Finally, a four-arm K-means log-spiral fit to the brightness-temperature ridge in $(\ln R, \phi)$ recovers pitch angles $i \simeq 5^\circ\text{--}15^\circ$ across clusters B–D, straddling the Sa/Sb Hubble–de Vaucouleurs boundary and consistent with the parallax-anchored M. J. Reid et al. (2014) multi-arm pitch range.

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